EDGE INTELLIGENCE LAB - 5

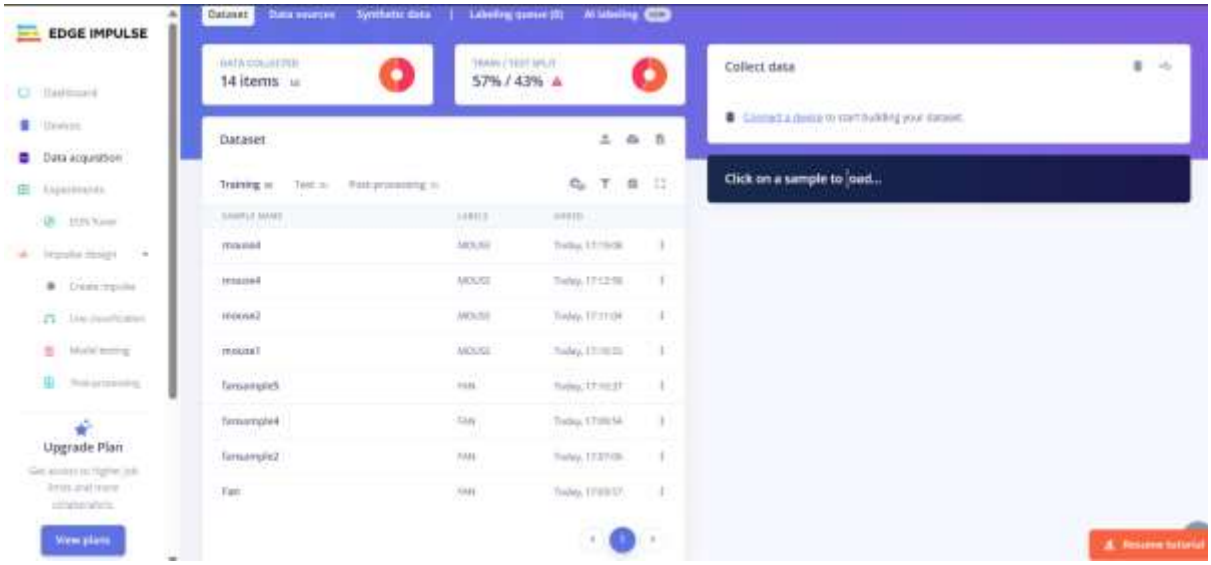
25MML0039

VISHWA M

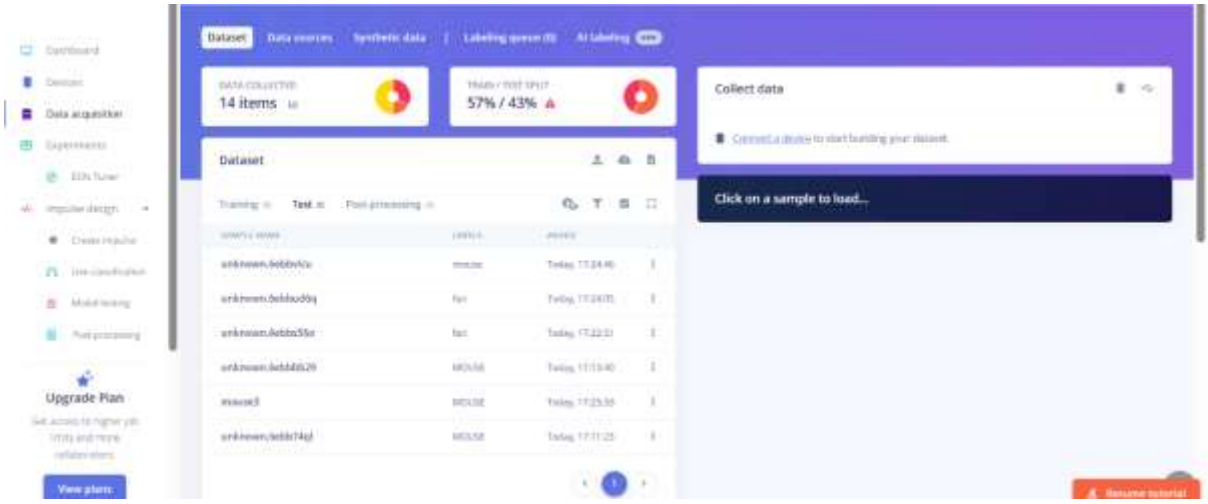
STEPS:

1. Create an account on Edge Impulse.
2. Login to the Edge Impulse platform.
3. Create a new project.
4. Click Data Acquisition.
5. Select Connect to device.
6. Choose Use mobile phone and scan the QR code.
7. Connect the mobile phone as a camera.
8. Capture sample images.
9. Collect images for training and testing.
10. Rename images with meaningful names.
11. Label all images correctly.

Training -8

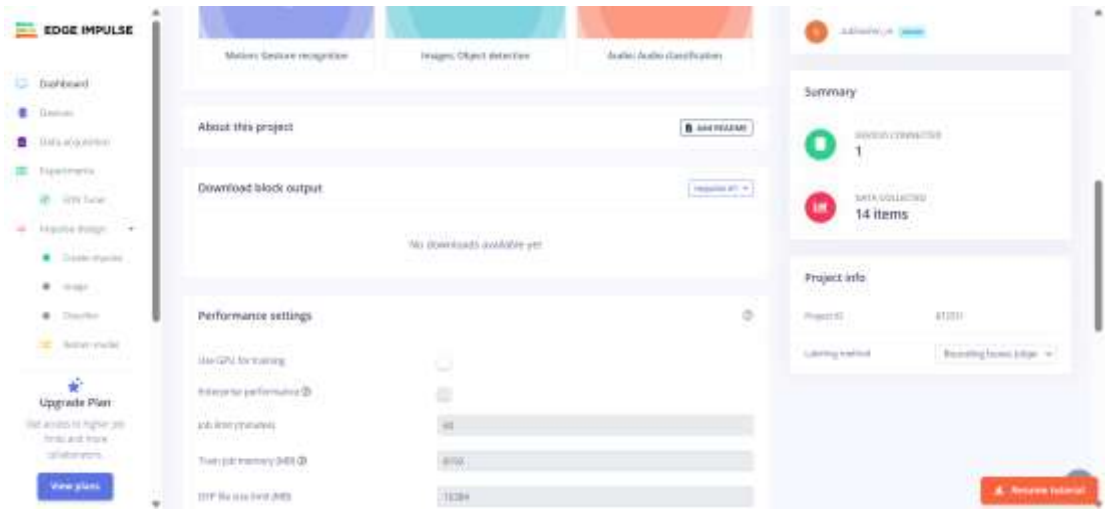


Testing-6

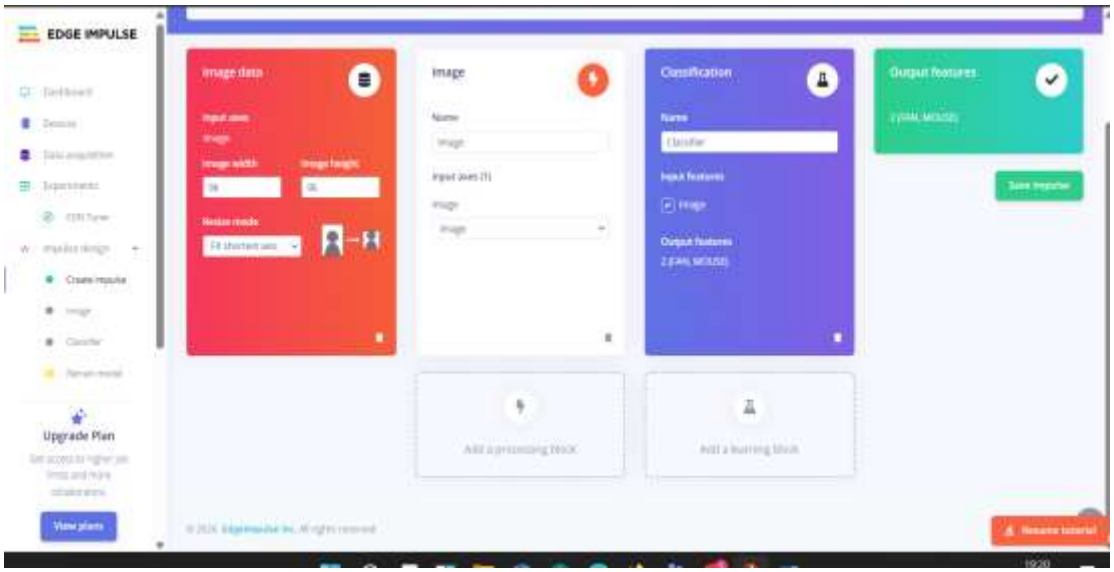


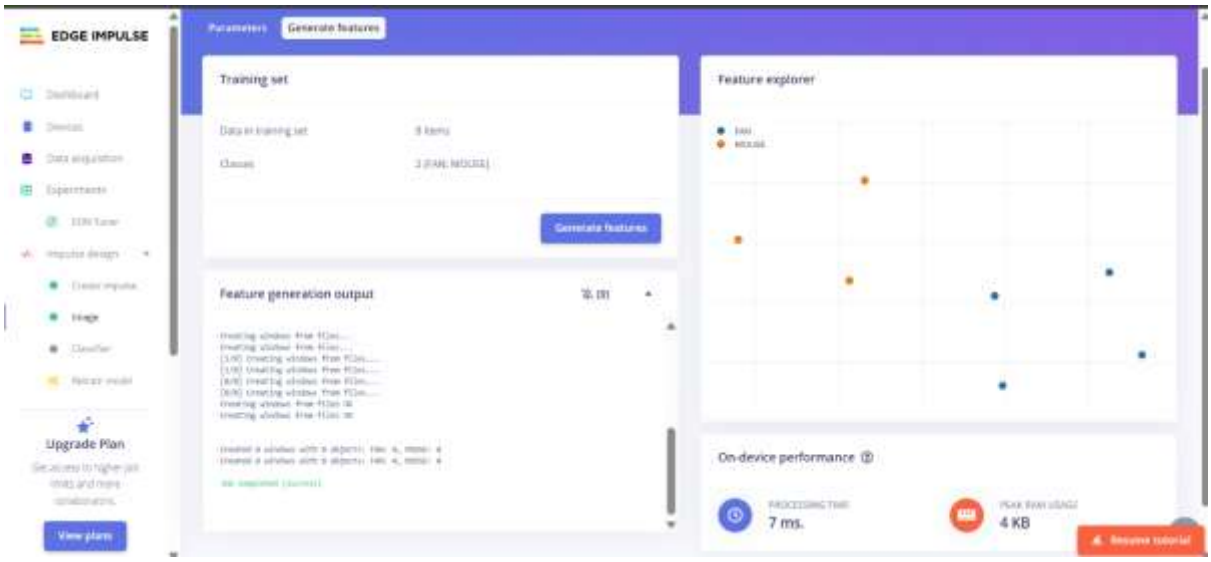
Steps to be performed:

1. Open **Impulse Design** and click **Create Impulse**.
2. Add a **Processing Block** as **Image**.
3. Add a **Learning Block** as **Image Classification**.
4. Click **Save Impulse**.
5. In **Impulse Design**, click on **Image**.
6. Click **Generate Features** to extract image features.
7. After feature generation is complete, go to **Image Classification** (Classifier).
8. Click **Start Training**.
9. Once training is finished, the **model accuracy** will be displayed for the images you uploaded.

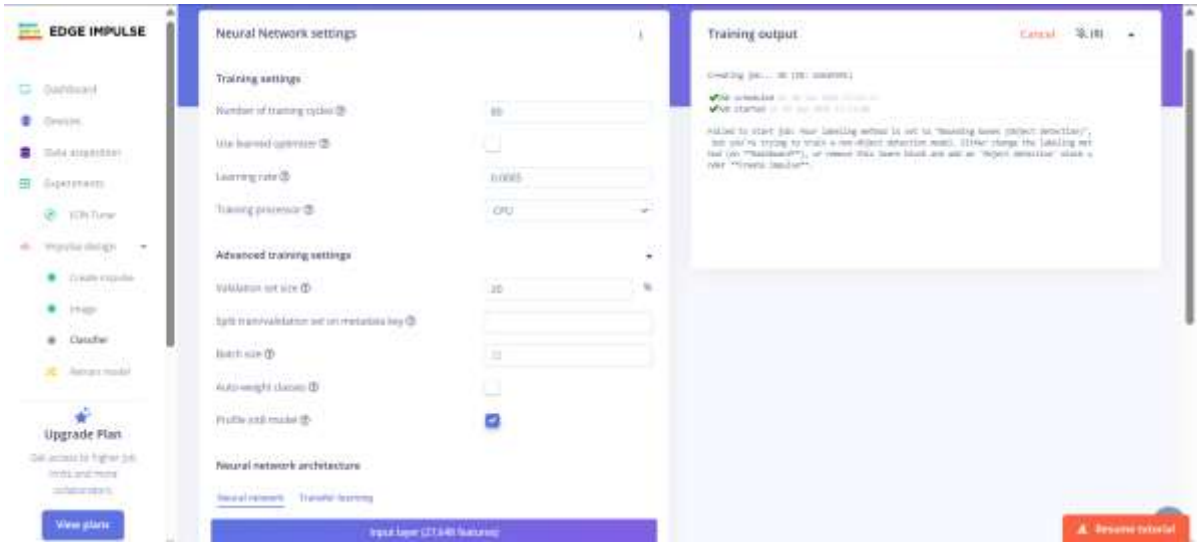


CREATE IMPLUSE





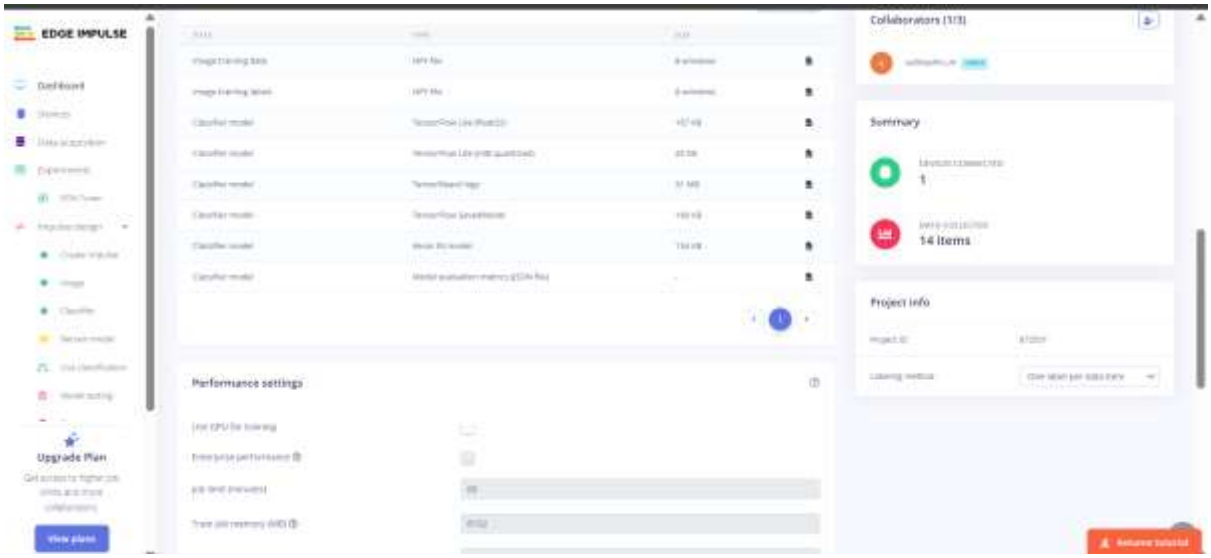
CLASSIFICATION



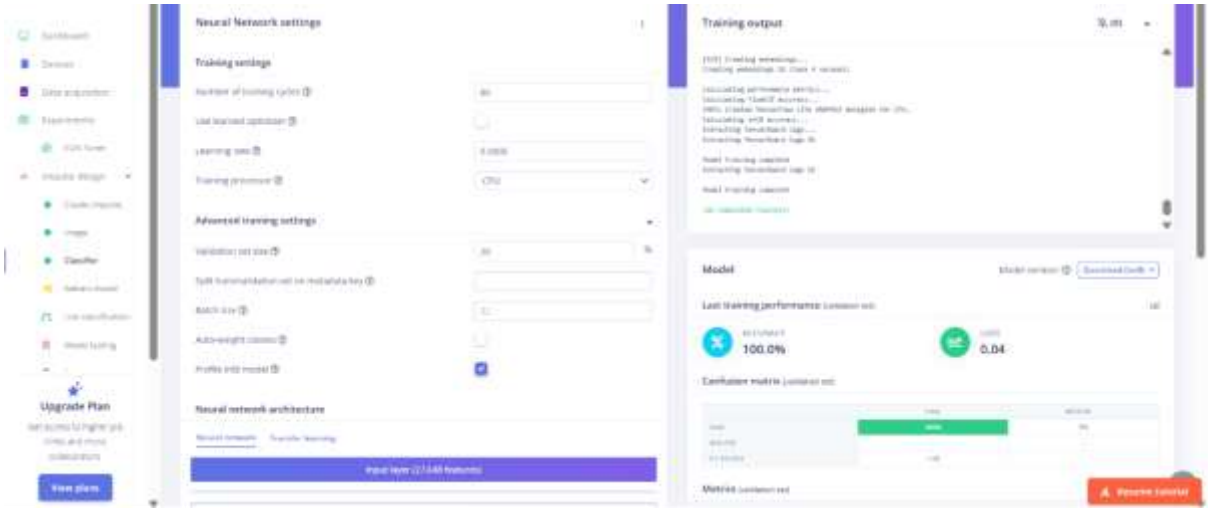
Error Fix and Image Classification:

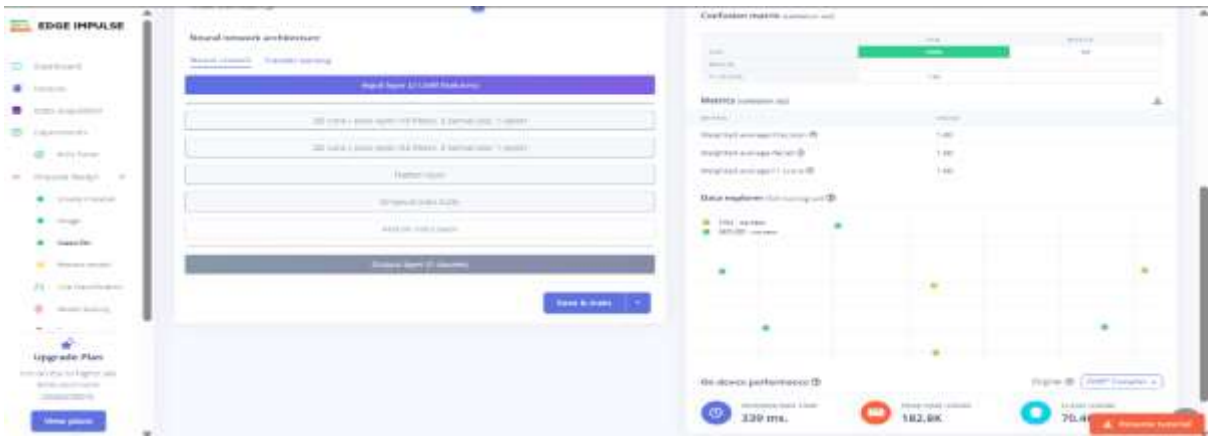
1. An error occurred because Bounding box labeling was used with an Image Classification model.
2. To fix this, go to Dashboard and change the labeling method to One label per data item.
3. Properly relabel all images using single class labels (no bounding boxes).
4. In Impulse Design, add Image as the processing block and Image Classification as the learning block.
5. Generate features, train the model, and the classifier achieved 100% accuracy.

SUMMARY


EDGE IMPULSE
Dashboard
Upgrade Plan
View plans
Collaborators (1/3)
Summary
1
14 items
Project info
Performance settings

CLASSIFIER


Upgrade Plan
View plans
Neural Network settings
Training settings
Advanced training settings
Neural network architecture
Training output
Model
100.0%
0.04


EDGE IMPULSE
Upgrade Plan
View plans
Neural network architecture
Confusion matrix
Matrix
Data explorer
On-device performance
339 ms.
182.8K